

Reusable-Vehicle Flight Clearance

From guided-wave SHM to a go/no-go decision with uncertainty

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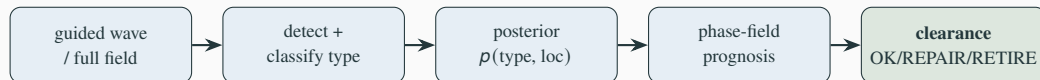
H3 fairing SHM so far: **detect & localise** debonding from guided waves (SAGE F1 0.79, LGSTA 0.86).

For a **reusable** stage the question becomes:

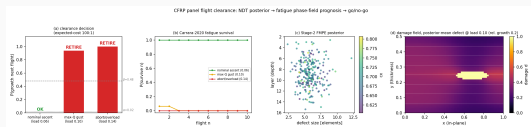
- is this fairing/stage OK to fly *again*?
- how many flights remain, with what confidence?

New layer

Turn the SHM posterior into a calibrated **flight-clearance decision** via phase-field crack prognosis.



- detect+classify: GNN on 9 sensor nodes $\times$ 24 features (SAGE F1 0.79, LGSTA 0.86); guided waves 16.5–300 kHz
- type/location posterior: long-term tags (13 damage states) + 5-class FEM localisation ($z \times \theta$ quadrants), *temperature-scaled, ECE-reported*
- prognosis: anisotropic AT2 + interface bands + Carrara-2020 fatigue $\Rightarrow P(\text{survive } n), \mathbb{E}[\text{flights}]$
- caveat: long-term tags carry a temporal confound (months differ in temperature & aging) — reported, not hidden



load	fatigue	$\mathbb{E}[\text{flights}]$	decision
nominal	ON	19.9	OK
nominal	off	11.0	OK
max-Q	ON	0.0	RETIRE

- $P(\text{survive } n)$ decays across flights once fatigue accumulates
- thresholds from expected cost, vehicle:repair = 100:1
 $\Rightarrow \alpha=0.02, \beta=0.48$

- **SpaceX practice, formalised:** condition-based monitoring + staged 20→40-flight certification + fleet-leader learning — made measurable and Bayesian, usable from the first article
- **Fleet learning (next):** hierarchical Bayes over tail numbers
- **Targets:** JAXA reusable-vehicle programmes; sim-to-real once measured full-field data arrives